

Monotonicity of Turán determinants for binary Krawtchouk polynomials

LESLIE P. POLZER

Independent Researcher

polzer@fastmail.com

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Abstract. Let $M, Q \geq 0$ be integers, set $D = M + Q$, and define $p_s = [w^s](1 + w)^M(1 - w)^Q$, with $p_s = 0$ outside $0 \leq s \leq D$. These are the binary Krawtchouk values $K_s(Q; D)$ in the generating-function normalization. We prove that the Turán determinants $\text{Tur}(s) = p_s^2 - p_{s-1}p_{s+1}$ satisfy

$$\text{Tur}(s) \geq \text{Tur}(s+1) \quad ([D/2] \leq s \leq D).$$

Consequently, the determinants form a symmetric weakly unimodal sequence and satisfy $\text{Tur}(s) \geq 1$ for every $0 \leq s \leq D$. By the dual Jacobi–Trudi identity, they are the rectangular Schur values $s_{(2^s)}(1^{\overline{M}}, (-1)^{\overline{Q}})$; for $D \geq 1$, this gives positivity and unimodality for a family of general linear character values at involutions. The proof combines recurrence comparisons in infinite regions, exact symbolic certificates for finitely many parameter families, and an exhaustive integer-arithmetic computation for the remaining finite cases. Source programs and deterministic replay logs are available in the accompanying repository.

Keywords. Krawtchouk polynomials, Turán determinant, Schur function, character value, unimodality, three-term recurrence, exact computation

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1. Introduction and main theorem

1.1. The theorem

Fix integers $M, Q \geq 0$, set $D = M + Q$, and define the integer sequence

$$p_s = [w^s](1 + w)^M(1 - w)^Q, \quad s = 0, 1, \dots, D, \quad (1.1)$$

with $p_s = 0$ outside $0 \leq s \leq D$. These are binary Krawtchouk values: with the standard generating function $\sum_k K_k(x; N) w^k = (1 + w)^{N-x}(1 - w)^x$ one has $p_s = K_s(Q; D)$, the degree- s binary (symmetric, $p = \frac{1}{2}$) Krawtchouk polynomial of length D evaluated at the integer point Q . Define the *Turán determinant* and the *Turán step* of this sequence,

$$\begin{aligned} \text{Tur}(s) &= p_s^2 - p_{s-1}p_{s+1}, \\ T_s &= \text{Tur}(s) - \text{Tur}(s+1) = p_s^2 + p_s p_{s+2} - p_{s+1}^2 - p_{s-1}p_{s+1}. \end{aligned} \quad (1.2)$$

Theorem 1.1 (*Main Theorem*). *For all integers $M, Q \geq 0$ and s with $\lceil D/2 \rceil \leq s \leq D$,*

$$T_s \geq 0.$$

Equivalently, $s \mapsto \text{Tur}(s)$ is non-increasing on the top half. Since $p_{D-s} = (-1)^Q p_s$, the sequence $\text{Tur}(s)$ is symmetric about $D/2$ and therefore weakly unimodal.

The following two corollaries are immediate by telescoping from the top end ($p_D = (-1)^Q$, $p_{D+1} = 0$, so $\text{Tur}(D) = 1$).

Corollary 1.2. *For all $M, Q \geq 0$:*

1. $\text{Tur}(s) \geq 1$ for every $0 \leq s \leq D$, with sharp constant 1;
2. $\text{Tur}(s) \leq \text{Tur}(\lceil D/2 \rceil)$ for $0 \leq s \leq D$.

Corollary 1.3 (*adjacent coefficient lattices*). *For $0 \leq s \leq D$, the two adjacent coefficient vectors*

$$v_s = (p_{s-1}, p_s), \quad v_{s+1} = (p_s, p_{s+1})$$

span a full-rank sublattice $L_s \subseteq \mathbb{Z}^2$ of index $[\mathbb{Z}^2 : L_s] = \text{Tur}(s)$. These lattice indices are symmetric and weakly increase from either endpoint toward the center.

Proof. The determinant of the ordered column matrix is $p_{s-1}p_{s+1} - p_s^2 = -\text{Tur}(s)$. Hence its absolute determinant, and therefore the lattice index, equals $\text{Tur}(s)$ by Corollary 1.2. The asserted ordering is the Main Theorem and reflection. ■

For example, take $(M, Q) = (5, 2)$. Then

$$(p_s)_{s=0}^7 = (1, 3, 1, -5, -5, 1, 3, 1), \quad (\text{Tur}(s))_{s=0}^7 = (1, 8, 16, 30, 30, 16, 8, 1).$$

The coefficients oscillate, while their Turán determinants remain positive, symmetric, and ordered toward the center.

Classical Turán inequalities concern the sign of $P_n(x)^2 - P_{n-1}(x)P_{n+1}(x)$ for a fixed degree index n and an argument x in the orthogonality interval [Sze48]. Theorem 1.1 instead concerns monotonicity in the degree index (equivalently, the generating-function coefficient index) at a fixed integer argument. The inequality $T_s \geq 0$ can fail when the integer argument Q is replaced by a real q . More precisely, for $q \in [0, D]$ define the coefficients by the same formal generating series

$$[w^s](1+w)^{D-q}(1-w)^q,$$

then at $(D, q, s) = (4, \frac{8}{7}, 2)$ one obtains $T_2 = -1430/117649 < 0$.

Closely related inequalities for binary Krawtchouk polynomials, their zeros, and their Christoffel–Darboux kernels were studied by Krasikov and Litsyn [KL96] and Krasikov [Kra01, Kra03, Kra05]. In particular, the inequality recorded in [Kra05, Theorem 5, (18), p. 79] couples adjacent Turán determinants in an exponential-coefficient normalization. Theorem 1.1 orders the determinants in the ordinary coefficient normalization at a fixed integer argument. Section 1.4 explains the role of normalization, and Section 8 gives a detailed comparison with these results and the character literature.

1.2. Schur-character interpretation

Let

$$z = (\underbrace{1, \dots, 1}_M, \underbrace{-1, \dots, -1}_Q),$$

and write $e_r(z)$ and $s_\lambda(z)$ for the elementary and Schur symmetric functions, respectively. As usual, (2^s) denotes the rectangular partition with s rows of length 2.

Proposition 1.4 (*Schur-character interpretation*). *For every $0 \leq s \leq D$,*

$$\text{Tur}(s) = s_{(2^s)}(z). \quad (1.3)$$

When $D \geq 1$, this is the irreducible polynomial-character value of $\text{GL}_D(\mathbb{C})$

$$\chi_{(2^s)}(\text{diag}(I_M, -I_Q)).$$

The rectangle-complement identity gives $s_{(2^s)}(z) = s_{(2^{D-s})}(z)$ independently of the Main Theorem.

Proof. The generating function in (1.1) is $\prod_{i=1}^D (1 + z_i w)$, so $p_r = e_r(z)$. Since the conjugate of (2^s) is (s, s) , the dual Jacobi–Trudi identity [Mac95, Chapter I, (3.5), p. 41] gives

$$s_{(2^s)}(z) = \det \begin{pmatrix} e_s(z) & e_{s+1}(z) \\ e_{s-1}(z) & e_s(z) \end{pmatrix} = p_s^2 - p_{s-1}p_{s+1}.$$

Schur polynomials are the irreducible polynomial characters of the general linear group; see [Mac95, Chapter I, Appendix A, (8.2) and (8.4), pp. 162–163]. Finally, for a partition λ inside the $2 \times D$ rectangle, with rectangle complement λ^c ,

$$s_\lambda(z_1^{-1}, \dots, z_D^{-1}) = (z_1 \cdots z_D)^{-2} s_{\lambda^c}(z_1, \dots, z_D).$$

Here $z_i^{-1} = z_i$, $(z_1 \cdots z_D)^2 = 1$, and the complement of (2^s) is (2^{D-s}) , proving the symmetry. ■

Thus Theorem 1.1 and Corollary 1.2 say that the rectangular character values in (1.3) are positive integers and form a symmetric weakly unimodal sequence as the rectangle height s varies. For $Q > 0$, the diagonal matrix has negative eigenvalues, so ordinary Schur positivity does not by itself determine the sign. The character interpretation identifies a canonical representation-theoretic object and suggests seeking a cancellation-free branching or restriction interpretation of the ordered differences T_s .

1.3. Adjacent minors and lattice indices

For a fixed integer argument Q , the values $(p_s)_{s=0}^D$ form a row of the binary Krawtchouk matrix. The Toeplitz matrix formed from this coefficient row has adjacent 2×2 minors of the form

$$\text{Tur}(s) = \det \begin{pmatrix} p_s & p_{s+1} \\ p_{s-1} & p_s \end{pmatrix}.$$

Corollary 1.3 identifies these minors with the indices of the lattices generated by successive coefficient vectors. This connects the theorem with minor and kernel inequalities in Hamming-scheme spectral analysis. The generating function, standard recurrences, and normalizations for Krawtchouk polynomials may be found in [KLS10, Section 9.11].

1.4. Normalization

The normalization in (1.1) is part of the statement. If $q_s = c_s p_s$, then

$$q_s^2 - q_{s-1}q_{s+1} = c_s^2 p_s^2 - c_{s-1}c_{s+1}p_{s-1}p_{s+1}.$$

Thus a Turán determinant is merely rescaled by a positive factor for every sequence p only when $c_s^2 = c_{s-1}c_{s+1}$, that is, when (c_s) is geometric. Degree-dependent orthonormal or endpoint normalizations of Krawtchouk polynomials therefore do not preserve Theorem 1.1.

The coefficient normalization is natural here because it is integral at the integer arguments under consideration, it has the exact reflection $p_{D-s} = (-1)^Q p_s$, and it fixes the endpoint $\text{Tur}(D) = p_D^2 = 1$. The last two features turn top-half monotonicity into symmetric weak unimodality and the sharp lower bound of Corollary 1.2. Section 8 compares this statement with Turán inequalities in other normalizations.

1.5. Guide to the proof

The top half $\{2s \geq D\}$ is divided into the bulk, flow, center, and gap regions. A positive-semidefinite quadratic form treats the bulk, and a Riccati comparison treats the flow region. The remaining gap is handled by fixed-offset certificates, fixed-argument certificates, and two parity-dependent recurrence arguments. Figure 1 and Table 1 summarize the decomposition; the proof is assembled in Section 6.2.

We assume $x = M - Q \geq 0$ without loss of generality. Indeed, T_s is invariant under $M \leftrightarrow Q$: the substitution $w \mapsto -w$ sends p_s to $(-1)^s p_s$, and every product in T_s has even index sum.

1.6. Computer-assisted components

Three kinds of argument occur below. First, the arguments for the bulk, center, flow region, and gap cells with even Q are given in the text; their accompanying programs replay algebraic identities and finite diagnostics but do not replace the written implications. Second, the finite-offset families, the fixed-argument families, and one polynomial-sign lemma for gap cells with odd Q are computer-assisted results: the text states their exact inputs, domains, and acceptance conditions, while the source programs construct and check the finite polynomial obligations. Third, Proposition 2.2 is an exhaustive computation in integer arithmetic over a declared finite set.

Every proof decision in the programs uses integer or rational arithmetic. A second C++ implementation checks the decisive exported polynomial witnesses and repeats the finite scan using a different coefficient construction. Section 7 specifies the certificate rules, software dependencies, and theorem-by-theorem inventory.

2. Notation and recurrence

2.1. Notation

Throughout:

$$\begin{aligned}
M, Q &\in \mathbb{Z}_{\geq 0}; & D &= M + Q; & x &= M - Q \geq 0; \\
p_s &= [w^s] (1+w)^M (1-w)^Q \text{ (integers; } p_s = 0 \text{ off } [0, D]); \\
s &\text{ top-half index, } D/2 \leq s \leq D; & u &= 2s - D \geq 0 \text{ (top-half offset);} \\
U &= (u+1)^2; & E &= (D+3)^2; \\
\text{Tur}(s) &= p_s^2 - p_{s-1}p_{s+1}; & T_s &= \text{Tur}(s) - \text{Tur}(s+1); \\
k &= D - s \text{ (distance from the top } = (D - u)/2).
\end{aligned}$$

Reflection: $p_{D-j} = (-1)^Q p_j$ for all j (substitute $w \rightarrow 1/w$). The top initial values are $p_D = (-1)^Q$ and $p_{D-1} = (-1)^Q x$. Logarithmic differentiation of the generating function gives the step-1 recurrence

$$(s+1)p_{s+1} = x p_s + (s-1-D)p_{s-1}. \quad (2.1)$$

Lemma 2.1 (*no consecutive zeros*). For $0 \leq j \leq D$, the two consecutive values p_j, p_{j+1} cannot both vanish.

Proof. If $j = 0$, the assumption contradicts $p_0 = 1$. If $1 \leq j \leq D$ and $p_j = p_{j+1} = 0$, then (2.1) at index j gives $p_{j-1} = 0$ because $j-1-D \neq 0$. Iteration gives $p_0 = 0$, again a contradiction. ■

We write $\varepsilon = (-1)^Q$ and $\binom{n}{k}$ for binomial coefficients. In the gap sections (Sections 4–6), we relabel the exponents when necessary so that $0 \leq Q \leq M$; thus Q denotes the smaller exponent.

2.2. Finite verification

A *cell* is an integer triple (M, Q, s) with $M, Q \geq 0$ and $\lceil (M+Q)/2 \rceil \leq s \leq M+Q$. We also express cells in the coordinates (D, x, u) defined above; these satisfy $u \equiv x \equiv D \pmod{2}$.

Proposition 2.2 (*finite verification*). The inequality $T_s \geq 0$ holds for every $M, Q \geq 0$ with $D \leq 1200$ and every $\lceil D/2 \rceil \leq s \leq D$.

Proof. The recurrence (2.1), initialized by $p_0 = 1$ and $p_1 = x$, was evaluated with GMP integers for all such triples. The computation examined 289,261,901 cells and found no negative value. Exact division was checked before every recurrence step. The recurrence-built values were checked against the defining binomial convolution for every coefficient with $D \leq 12$ and at deterministic indices in six larger rows, reaching $D = 1200$. The program, output, and hash are listed in Section 7. ■

A separate Python computation verifies the recurrence for $M, Q \leq 30$ and checks $T_s \geq 0$ directly for $M, Q \leq 90$.

3. Quadratic form and regime decomposition

3.1. The bulk band

Theorem 3.1 (*quadratic-form identity and the bulk band*). *Eliminating p_{s+2} and p_{s-1} from T_s by the step-1 recurrence (2.1) gives the exact identity*

$$\begin{aligned} T_s &= A p_s^2 + B p_s p_{s+1} + C p_{s+1}^2, \\ A &= \frac{u+2}{s+2}, \quad B = \frac{-x(u+1)}{(D+1-s)(s+2)}, \quad C = \frac{u}{D+1-s}, \end{aligned} \quad (3.1)$$

with negated discriminant

$$4AC - B^2 = \frac{u(u+2)((D+3)^2 - (u+1)^2) - x^2(u+1)^2}{((D+1-s)(s+2))^2}. \quad (3.2)$$

Hence for $u \geq 1$, writing $U = (u+1)^2$, $E = (D+3)^2$ (note $u(u+2) = U - 1$):

$$x^2 U \leq (U - 1)(E - U) \implies T_s \geq 0 \quad (3.3)$$

without further information about the sequence. The condition defines the bulk band $U \in [U_-, U_+]$, where

$$U_{\pm} = \frac{h \pm \sqrt{h^2 - 4E}}{2}, \quad h := E + 1 - x^2,$$

are the roots of $U^2 - hU + E = 0$.

Proof. Substitute (2.1) for p_{s+2} and p_{s-1} in (1.2); simplification gives (3.1) and (3.2). The displayed roots are real throughout the parameter domain: since $0 \leq x \leq D$,

$$h = E + 1 - x^2 \geq (D+3)^2 + 1 - D^2 = 6D + 10 > 2(D+3) = 2\sqrt{E}.$$

If $u \geq 1$, then $A, C > 0$, and the quadratic form is positive semidefinite whenever $4AC - B^2 \geq 0$. ■

Remark (sharpness). The *sign* of the negated discriminant is unchanged under the indicated basis changes: expressing T_s in a shifted recurrence basis multiplies the negated discriminant by a positive square. For example, the negated discriminant in the (p_{s-2}, p_{s-1}) basis is the displayed one times

$$\left(\frac{(D+1-s)(D+2-s)}{s(s+1)} \right)^2.$$

Thus no change among the shifted recurrence bases makes the form positive semidefinite outside the bulk band. The remaining arguments therefore use additional information from the Krawtchouk recurrence.

3.2. The center

Theorem 3.2 (*central identity, $u = 0$, D even*). By reflection ($p_{D-j} = (-1)^Q p_j$): for Q odd, $p_{D/2} = 0$ and $T_{D/2} = 0$ identically. For Q even, the reflection constraint $p_{s-1} = p_{s+1}$ at $s = D/2$ collapses the recurrence to the rank-one relation $(D+2)p_{s+1} = xp_s$, and

$$T_{D/2} = \frac{4p_{D/2}^2((D+2)^2 - x^2)}{(D+2)^2(D+4)} \geq 0, \quad (3.4)$$

with equality if and only if $p_{D/2} = 0$.

Proof. If Q is odd, reflection gives $p_{D/2} = 0$ and $p_{D/2-1} = -p_{D/2+1}$, which makes the four terms in $T_{D/2}$ cancel. If Q is even, reflection gives $p_{s-1} = p_{s+1}$ at $s = D/2$. Substitution in (2.1) yields $(D+2)p_{s+1} = xp_s$; inserting this relation into (1.2) and using the recurrence once more for p_{s+2} gives (3.4). Its remaining factor is strictly positive because

$$(D+2)^2 - x^2 \geq (D+2)^2 - D^2 = 4D + 4 > 0. \quad \blacksquare$$

The recurrence arguments in Sections 5 and 6 start from reflection constraints at the center.

When M and Q are both odd, D is even and $T_{D/2} = 0$. This completely describes equality at the central step. If D is even, then M, Q have the same parity. The odd-odd case is the first case of Theorem 3.2; in the even-even case, $p_{D/2} = 0$ together with $(D+2)p_{D/2+1} = xp_{D/2}$ would give two consecutive zeros, contrary to Lemma 2.1. We do not claim a classification of equality at noncentral steps.

3.3. The flow region: Riccati comparison

Theorem 3.3 (*flow region*). Let $k = D - s \geq 1$. Whenever its radicand is nonnegative, let

$$w_s = \sqrt{x^2 - 4k(D+1-k)}$$

denote the nonnegative square root. On the flow region $w_s^2 \geq 0$, equivalently $(u+1)^2 \geq (D+1)^2 - x^2$ by the identity $(D+1)^2 - (u+1)^2 = 4(s+1)(D-s)$, one has $T_s \geq 0$.

Proof. The flow region with $k \geq 1$ is empty when $x = 0$, so assume $x > 0$. On the top half, $0 \leq k \leq D/2$, and $k \mapsto 4k(D+1-k)$ is increasing. Hence if the target index is in the flow region, so is every index between it and $D-1$; the following downward induction runs on this connected tail. The center $u = 0$ cannot lie in the flow region, because there $4k(D+1-k) = D(D+2) > D^2 \geq x^2$. Thus $u \geq 1$ and the quadratic-form coefficient C is positive throughout the tail.

Use the ratio variable $r_s = p_{s+1}/p_s$. The top initial values $p_D = (-1)^Q$, $p_{D-1} = (-1)^Q x$ give $r_{D-1} = 1/x$, and the recurrence gives the downward Möbius flow

$$r_{s-1} = \frac{D+1-s}{x - (s+1)r_s}.$$

Set $\phi_s = (x + w_s)/2$ and define the auxiliary downward Riccati flow $y_{s-1} = x - (s+1)(D-s)/y_s$, $y_{D-1} = x$. Then:

1. (*comparison*) The inequality $\phi_{s-1} \leq x - (s+1)(D-s)/\phi_s$ reduces, after clearing denominators by $4(s+1)(D-s) = x^2 - w_s^2$, to $(w_s - w_{s-1})(x + w_s) \geq 0$. This holds because $w_s^2 - w_{s-1}^2 = 4u > 0$. Induction from $y_{D-1} = x \geq \phi_{D-1}$ gives $y_s \geq \phi_s > 0$ on the connected tail.

Starting from $r_{D-1} = 1/x = (D - (D-1))/y_{D-1}$, a simultaneous induction in the two displayed recurrences gives $r_s = (D-s)/y_s = k/y_s$. Consequently $0 < r_s \leq k/\phi_s$, every Möbius denominator is positive, and no p_s in the tail vanishes.

2. (*window value*) The quadratic-form value of Theorem 3.1 at $r = k/\phi_s$ is $I_1/((s+2)(D+1-s)\phi_s^2)$ with

$$\begin{aligned} I_1 &= (u+2)(k+1)\phi_s^2 - x(u+1)k\phi_s + u(D-k+2)k^2 = P_1 + Q_1w_s, \\ Q_1 &= \frac{1}{2}x(D-k+2) \geq 0, \\ P_1 &= \frac{1}{2}[x^2(D+2-k) - 2k(D^2 - 2Dk + 3D + 2k^2 - 2k + 2)], \end{aligned} \quad (3.5)$$

The coefficient $D+2-k$ of x^2 is positive. Thus P_1 is increasing in x^2 , and at the flow boundary $x^2 = 4k(D+1-k)$ it equals $k(D+2)(u+1) > 0$. Since also $Q_1, w_s \geq 0$, this proves $I_1 \geq 0$ throughout the flow region.

3. (*window vertex*) Since $C > 0$, the inequality $k/\phi_s \leq -B/(2C)$ reduces to

$$I_2 = x(u+1)\phi_s - 2u(D-k+2)k = P_2 + Q_2w_s, \quad (3.6)$$

where

$$Q_2 = \frac{x(u+1)}{2}, \quad P_2 = \frac{x^2(u+1) - 4k(D^2 - 3Dk + 2D + 2k^2 - 4k)}{2}.$$

Since $u+1 > 0$, P_2 is increasing in x^2 ; at the flow boundary it equals $2k(k+1) > 0$. Hence $I_2 \geq 0$.

4. Since $A, C \geq 0 \geq B$, the quadratic $A + Br + Cr^2$ is decreasing on $[0, -B/(2C)]$; by (1)–(3), $T_s/p_s^2 = A + Br_s + Cr_s^2 \geq (\text{value at } k/\phi_s) \geq 0$. ■

3.4. Coverage and the gap

Corollary 3.4 (*coverage*). *For $D = 0$ the only cell is $s = 0$ and $T_0 = 1$. For $D \geq 1$, every top-half cell is covered by the top, center, bulk, or flow arguments except the set*

$$\mathcal{G} = \{u \geq 1 : x^2U > (U-1)(E-U), \quad U < (D+1)^2 - x^2\}, \quad (3.7)$$

equivalently $\{u \geq 1 : U < U_-\}$.

Proof. At the flow boundary $U_0 = (D+1)^2 - x^2$ the bulk margin is

$$(U_0 - 1)(4D + 8) - x^2 = (D^2 + 2D - x^2)(4D + 8) - x^2 > 0$$

for $0 \leq x \leq D$ (decreasing linearly with x^2 , with value $7D^2 + 16D$ at $x^2 = D^2$). Hence $U_- \leq U_0 \leq U_+$, and every top-half cell is covered by $\{s = D\}$ ($T_D = 1$), $\{u = 0\}$ (Theorem 3.2), the bulk ($U_- \leq U \leq U_0$), and the flow ($U \geq U_0$), except for (3.7). ■

Lemma 3.5 (*heredity of the gap*). *If (D, x, u) is a gap cell and $u' \equiv u \pmod{2}$ satisfies $1 \leq u' \leq u$, then (D, x, u') is also a gap cell.*

Proof. Corollary 3.4 identifies the gap exactly with $1 \leq (u+1)^2 < U_-$. Since $(u'+1)^2 \leq (u+1)^2$, the same inequality holds for u' . The parity condition makes $s' = (D+u')/2$ an integer index. ■

Using $E - x^2 = (2Q+3)(2M+3)$, failure of the bulk inequality is equivalent to

$$(2Q+3)(2M+3) < U + \frac{E}{U} - 1. \quad (3.8)$$

Together with the non-flow condition in (3.7), this characterizes the gap. Thus gap cells occur only when the two exponents are unequal. An exact census shows that the first gap cell with $u \geq 4$ and $Q \geq 3$ appears at $D = 442$, $(Q, u) = (3, 4)$. Over all Q the first such cell is instead $(D, Q, u) = (142, 0, 4)$; it is covered by Proposition 4.3 before this residual region is considered.

The **regime-decomposition** program also checks this partition for every top-half cell with $D \leq 90$.

Lemma 3.6 (*gap strip*). *After relabeling so that $Q \leq M$, every gap cell satisfies*

$$(u+1)^2 = U < U_- \leq \frac{7}{6} \frac{D+3}{2Q+3}. \quad (3.9)$$

Proof. Here

$$h = E + 1 - x^2 = (2Q+3)(2M+3) + 1,$$

and $h - 3(D+3) = 4MQ + 3M + 3Q + 1 > 0$. Thus $h > 3\sqrt{E}$. Since

$$U_- = \frac{2E}{h + \sqrt{h^2 - 4E}},$$

putting $t = h/\sqrt{E}$ shows that

$$\frac{U_-}{E/h} = \frac{2t}{t + \sqrt{t^2 - 4}}.$$

The last function is decreasing for $t \geq 2$, and its value at $t = 3$ is $6/(3 + \sqrt{5}) \leq 7/6$. Hence $U_- \leq (7/6)E/h$. Finally $h \geq (2Q+3)(D+3)$ because $2M+3 \geq D+3$ when $Q \leq M$. ■

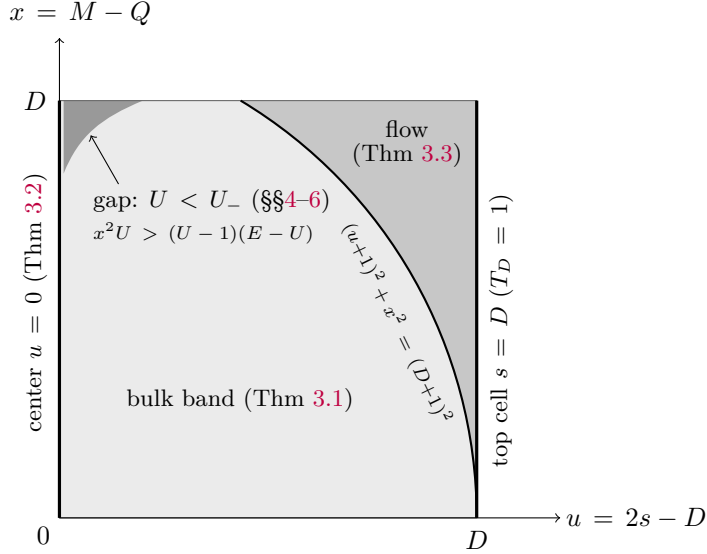


Figure 1. Coverage of the top half $\{2s \geq D\}$ in the (u, x) -plane for fixed D (schematic; lattice cells satisfy $u \equiv x \equiv D \pmod{2}$). For this schematic, the overlap of the bulk and flow regions is assigned to the flow region, beyond the displayed circle. The dark gap is removed from the light bulk shading: it is the part of the non-flow region where the bulk quadratic form is indefinite, $\{u \geq 1 : U < U_-\}$. It is treated in Sections 4–6.

3.5. Coverage and thresholds

Table 1 records which statement covers each region. Theorems 3.1, 3.2, 3.3, and 4.2, together with Propositions 4.1 and 4.3, cover every cell with $D \leq 4586$; the first cell not covered by those results is $(D, Q, u) = (4587, 3, 15)$. Corollary 4.6 extends the threshold to $D \leq 14826$, and Theorems 5.1 and 6.2 cover all remaining cells.

The numerical cutoffs are bookkeeping constants, not conjecturally sharp features of the inequality. The exact scan through $D = 1200$ absorbs the finite-offset nondegeneracy remainder $D < 1191$; the fixed-argument edge-flow onsets are smaller still (at most 93). In the final residual region, $x \leq D - 26$ is simply $x = D - 2Q$ with $Q \geq 13$.

Region or slice of the gap	Covered by
top cell $s = D$	endpoint identity $T_D = 1$
center column $u = 0$	Theorem 3.2
bulk band $x^2 U \leq (U - 1)(E - U)$	Theorem 3.1
flow region $U \geq (D + 1)^2 - x^2$	Theorem 3.3
coverage: everything else = gap	Corollary 3.4
gap, $u \leq 3$	Proposition 4.1
gap, $Q \geq 3, 4 \leq u \leq 14$	Theorem 4.2
gap, $Q \leq 2$	Proposition 4.3
gap, $3 \leq Q \leq 12$	Theorem 4.5
gap, $u \geq 15, Q \geq 13, Q$ even	Theorem 5.1
gap, $u \geq 15, Q \geq 13, Q$ odd	Theorem 6.2
all cells with $D \leq 1200$	Proposition 2.2

Table 1. Coverage of the proof. Throughout the table, $Q \leq M$.

4. Reflection collapse and finite families

The gap is next treated through two finite collections of parameter families: fixed-offset families (Proposition 4.1 and Theorem 4.2) and fixed-argument strips (Proposition 4.3 and Theorem 4.5).

4.1. Reflection collapse: gap offsets $u \leq 3$

Proposition 4.1 (*reflection collapse*). *Every gap cell with $u \leq 3$ has $T_s \geq 0$.*

Proof. For $s = (D + u)/2$ reflection gives $p_{s-u} = (-1)^Q p_s$. Combining this reflection constraint with the $(u + 2)$ -step recurrence chain determines the ratio p_{s-u+1}/p_{s-u} as an explicit rational function of (M, Q) and yields closed forms

$$T_s = R_u(M, Q) p_{s-u}^2.$$

For $u = 1, 2, 3$, with $\varepsilon = (-1)^Q$ indicated by the superscript, the recurrence gives

$$\begin{aligned}
R_1^+ &= \frac{32(M+2)(Q+1)}{(D+3)^2(D+5)}, & R_1^- &= \frac{32(M+1)(Q+2)}{(D+3)^2(D+5)}, \\
R_2^+ &= \frac{16(M+1)(Q+1)[(M-Q)^2 + 2M^2 + 2Q^2 + 6M + 6Q]}{x^2(D+4)^2(D+6)}, \\
R_2^- &= \frac{32(M+2)(Q+2)}{(D+4)^2(D+6)}, \\
R_3^+ &= \frac{64(M+2)(Q+1)[(M-Q)^2 + 4Q^2 + 12Q - 1]}{(M-3Q-1)^2(D+5)^2(D+7)}, \\
R_3^- &= \frac{64(M+1)(Q+2)[(M-Q)^2 + 4M^2 + 12M - 1]}{(3M-Q+1)^2(D+5)^2(D+7)}.
\end{aligned}$$

These expressions are nonnegative in the stated range: the two terms containing -1 are nonnegative for $Q \geq 1$ and $M \geq 1$, respectively, while for R_3^+ with $Q = 0$ one has $D = M \geq u = 3$ (a coefficient cell requires $u \leq D$), so $M^2 - 1 > 0$. The only potentially negative residual values have $D < u$ and are not coefficient cells. The collapse denominators do not vanish on the gap: $x = 0$ fails the gap inequality outright; $M = 3Q + 1$ gives $16x^2 - 15(E - 16) = -(176Q^2 + 416Q - 16) < 0$ for $Q \geq 1$; and $3M - Q + 1 > 0$ for $M \geq Q$. Moreover, the seed p_{s-u} cannot vanish: once the displayed collapse denominator is nonzero, the reflection constraint and the recurrence force $p_{s-u+1} = 0$ as well, contradicting Lemma 2.1. Thus the ratio elimination used above is legitimate, and every gap cell with $u \leq 3$ has $T_s \geq 0$. ■

4.2. Fraction-free constrained forms: gap offsets $4 \leq u \leq 14$

Theorem 4.2 (*finite gap offsets*). *Every gap cell with $Q \geq 3$ and $4 \leq u \leq 14$ has $T_s \geq 0$.*

Proof. For each u and parity class $\varepsilon = (-1)^Q$, the reflection constraint is encoded fraction-free: with the scaled chain $P_i = c_i p_{k_0+i}$ ($k_0 = s - u = D - s$; $c_0 = c_1 = 1$ and $c_{i+1} = c_i(k_0 + i + 1)$ for $i \geq 1$), the recurrence becomes

$$P_{i+1} = xP_i + (k_0 + i - 1 - D) \frac{c_i}{c_{i-1}} P_{i-1}.$$

Thus every P_i has polynomial coefficients in the two seeds $(a, b) = (p_{k_0}, p_{k_0+1})$. All scale factors used below are positive: $k_0 \geq 0$ and every new factor $k_0 + i + 1$ is positive. In particular,

$$\tilde{T} := T_s c_{u+1} c_{u+2} = \mathcal{A}a^2 + \mathcal{B}ab + \mathcal{C}b^2$$

is a polynomial quadratic form. The constraint $p_s = \varepsilon p_{k_0}$ reads $\lambda a + \mu b = 0$ with polynomial λ, μ , and pure algebra gives the two identities

$$\tilde{T} \mu^2 = G a^2, \quad \tilde{T} \lambda^2 = G b^2, \quad G := \mathcal{A} \mu^2 - \mathcal{B} \lambda \mu + \mathcal{C} \lambda^2, \quad (4.1)$$

at every cell satisfying the constraint, without division. Hence $T_s \geq 0$ follows at a cell whenever $G \geq 0$ there and $(\lambda, \mu) \neq (0, 0)$. The **finite-gap-offsets** replay proves the following finite computer-assisted certificate for each (u, ε) :

1. (*sector*) $\text{gap}(u)$ with $Q \geq 3$ forces $Q < M/(u+1)^2$, by the estimate $2Q + 3 < (D + 3)/U + U/(D + 3)$ with $(D + 3)/U > 8$. Indeed, $U < U_- \leq D + 3$ makes $\tau = (D + 3)/U > 1$, and then $\tau + 1/\tau > 2Q + 3 \geq 9$ forces $\tau > (9 + \sqrt{77})/2 > 8$. The exact census already quoted in Section 3 (and repeated by this verifier) shows that no gap cell with $u \geq 4$ and $Q \geq 3$ has $D < 442$, so $Q < M/25$ gives $M > \frac{25}{26} D \geq \frac{25}{26} \cdot 442 = 425$. Since M is integral, this gives $M \geq 426$; all subsequent bounds require only $M \geq 425$.
2. (*two reduced constraints*) We call a chosen reflection constraint a *pinning*. For the primary pinning $p_s = \varepsilon p_{k_0}$, let $g_1 = \gcd(\lambda_1, \mu_1)$ and factor out the greatest common divisor by writing $(\lambda_1, \mu_1) = g_1(\lambda'_1, \mu'_1)$, so $G_1 = g_1^2 G'_1$. The same construction for

the alternate pinning $p_{s+1} = \varepsilon p_{k_0-1}$ gives (λ_2, μ_2) , g_2 , and G'_2 . The two fraction-free identities (4.1) hold for each reduced constraint wherever its g_r is nonzero.

3. (*positivity*) $G' \geq 0$ on the sector via factorization: even-multiplicity factors are non-negative; odd-multiplicity factors are positive because their leading homogeneous form $L(t)$ is positive on $[0, 1/(u+1)^2]$ (an exact 512-point partition and a derivative bound), while explicit weighted bounds control all lower-degree terms. This is done for both G'_1 and G'_2 . The resulting thresholds satisfy $M_0(u) \leq 153$ through $u = 14$, so the sector bound $M \geq 425$ automatically lies beyond every positivity threshold.
4. (*nondegeneracy*) For each of the two pinning, the reduced pair (λ'_r, μ'_r) has no common integer zero on the part of the sector with $M > 1145$ (resultant in Q , integer-root candidates, exact check). The omitted range is already covered by Proposition 2.2: since $Q < M/(u+1)^2 \leq M/25$, $M \leq 1145$ implies $D = M + Q < 1191 < 1200$.

Where the first common factor g_1 vanishes, we use the alternative reflection constraint $p_{s+1} = \varepsilon p_{k_0-1}$. A further resultant check proves that $\{g_1 = 0\} \cap \{g_2 = 0\}$ contains no integer sector point with $M > 1145$. Thus the second common factor is nonzero there, and the second reduced identities are valid. The second reduced pair is nonzero by its own nondegeneracy check above. Both $g_2 \neq 0$ and $(\lambda'_2, \mu'_2) \neq (0, 0)$ are used at this cell.

Here G' in item (3) denotes either reduced form G'_r . Consequently at least one reduced identity is valid at every cell, its right-hand side is nonnegative, and a nonzero reduced constraint coefficient forces $\tilde{T} \geq 0$. Positivity of the scale factors then gives $T_s \geq 0$. The recursive inputs, generic sign lemmas, factor degrees, domination thresholds, and resultant-candidate reporting are stated explicitly in Section 7.2. ■

A worked finite-offset certificate. We illustrate the certificate construction with $u = 4$ and $\varepsilon = +1$. After removing the common factor $g = M - Q = x$ from (λ, μ) , the program obtains

$$\begin{aligned}\lambda' &= -\frac{x(D+4)}{2}, \\ \mu' &= \frac{M^2 - 6MQ - 2M + Q^2 - 2Q}{2}, \\ G' &= \frac{(M+1)(Q+1)D^2(D+2)^2(D+4)^2}{128} F(M, Q),\end{aligned}$$

where

$$\begin{aligned}F(M, Q) &= 5M^4 - 20M^3Q + 78M^2Q^2 - 20MQ^3 + 5Q^4 \\ &\quad + 144M^2Q + 144MQ^2 - 20M^2 + 472MQ - 20Q^2.\end{aligned}$$

On the certified sector put $t = Q/M < 1/25$. Dropping positive terms gives

$$\frac{F(M, Q)}{M^4} \geq \frac{13121}{3125} - \frac{20(1+t^2)}{M^2} > 0 \quad (M \geq 425).$$

Moreover $x \neq 0$ on a gap cell, so $g \neq 0$ and $\lambda' \neq 0$; the alternate pinning and the resultant checks are therefore not needed for this family's argument (the verifier still performs them).

The identities (4.1) therefore prove the $u = 4, \varepsilon = +1$ family directly. The other 21 offset-parity families follow the same construction, with the longer factor and nondegeneracy data recorded by the certificate generator.

Together with the preceding results and Proposition 4.3, this covers every offset $u \leq 14$ and every cell with $D \leq 4586$.

4.3. Small-argument closed forms: $\min(M, Q) \leq 2$

Proposition 4.3 ($\min(M, Q) \leq 2$, all D). *If $\min(M, Q) \leq 2$, then $T_s \geq 0$ at every top-half index.*

Proof. Put $v := D - u = 2(D - s)$. For fixed Q , exact cancellation gives

$$\frac{T_s}{\binom{M}{s}^2} = \frac{N_Q(D, u)}{\Delta_Q(D, u)} \quad (u \leq D - 10),$$

where $\Delta_Q > 0$: after factorization it is a positive constant times squares, positive factors $D + u + c$, and the positive factor $D + 2 - u$. (Equivalently, the denominator before this sign adjustment has the single negative factor $u - D - 2$.) The edge $v < 10$ is covered by the trivial top cell when $s = D$, and by Theorem 3.3 for $D \geq 24$ when $s < D$. The oriented numerators N_Q are:

$$\begin{aligned} N_0 &= (D+1)(D+2)(1+u), \\ N_1 &= D(D+1)u(u+1)(u+2), \\ N_2 &= D(D-1)S_2, \\ S_2 &= c_0 + c_1u + c_2u^2 + c_3u^3 + c_4u^4 + c_5u^5 \\ &= 3D(D+2) + (D+2)(3D-4)u - 2(3D+4)u^2 \\ &\quad - 2(D-2)u^3 + 5u^4 + u^5. \end{aligned} \tag{4.2}$$

The $Q = 0$ numerator is positive; the $Q = 1$ numerator is nonnegative and vanishes exactly at $u = 0$ (Theorem 3.2, odd parity); and $S_2 \geq 0$ on the strip $u^2 \leq D/13$ by *paired domination*:

$$13c_0 + Dc_2 = D(33D + 70) \geq 0, \quad 13c_1 + Dc_3 = 37D^2 + 30D - 104 \geq 0,$$

with $c_4, c_5 \geq 0$ (the second paired polynomial is positive for $D \geq 2$, the only relevant $Q = 2$ range).

It remains to justify that this strip contains the whole $Q = 2$ gap. Here $x = D - 4$, $E = (D + 3)^2$, and $h = E + 1 - x^2 = 14D - 6$. Since $h \geq 2\sqrt{E}$ for $D \geq 3$, rationalizing the smaller root gives

$$U < U_- = \frac{2E}{h + \sqrt{h^2 - 4E}} \leq \frac{E}{h - \sqrt{E}} \leq \frac{(D+3)^2}{13D-9}.$$

Moreover

$$13(D+3)^2 \leq (D+13)(13D-9) \iff 82D - 234 \geq 0.$$

Consequently $(u+1)^2 \leq D/13 + 1$, and thus $u^2 = (u+1)^2 - 2u - 1 \leq D/13$ for $u \geq 0$. Off the strip Corollary 3.4 applies. The edge cells with $D < 24$ lie in Proposition 2.2. ■

4.4. Fixed-argument strips: $3 \leq Q \leq 12$

Lemma 4.4 (*fixed-argument certificate*). *Fix $3 \leq Q \leq 12$ and set*

$$R_j(M, s) := \sum_{b=0}^Q (-1)^b \binom{Q}{b} \begin{cases} \prod_{i=0}^{j-b-1} \frac{M-s-i}{s+i+1}, & j-b \geq 0, \\ \prod_{i=0}^{b-j-1} \frac{s-i}{M-s+i+1}, & j-b < 0. \end{cases}$$

On the strip $0 \leq u \leq D - 2Q - 6$,

$$\frac{T_s}{\binom{M}{s}^2} = R_0^2 + R_0 R_2 - R_1^2 - R_{-1} R_1.$$

After substituting $M = D - Q$, $s = (D + u)/2$, orienting the denominator, and removing positive content, write its numerator as $S_Q = \sum_j c_j(D) u^j$. On $0 \leq u \leq D - 2Q - 6$ the oriented denominator is positive. Every negative term $c_j u^j$ has the unique partner two degrees below, and for $D \geq 2Q + 6$

$$B_Q(D) := \frac{7}{6} \frac{D+3}{2Q+3},$$

$$c_j(D) \leq 0, \quad c_{j-2}(D) \geq 0, \quad c_{j-2}(D) + c_j(D) B_Q(D) \geq 0.$$

All unpaired coefficients are nonnegative on the same ray.

Proof. The displayed finite sum follows by dividing the defining binomial convolution for p_{s+j} by $\binom{M}{s}$. Its denominator factors as a positive constant times squares and positive factors $D + u + c$, together with $D + 2 - u > 0$ on the stated interval. Exact expansion supplies the $c_j(D)$. Their signs and the partner inequalities are univariate polynomial claims on $[2Q + 6, \infty)$ and are certified by integer Sturm chains. For even Q , the partner indices $j - 2$ are $4r, 4r + 1$ ($0 \leq r < Q/2$); for odd Q , they are $4r + 2, 4r + 3$ ($0 \leq r < (Q - 1)/2$). These lists exhaust the negative coefficients. The exact generation and replay specification is given in Section 7.2. ■

Theorem 4.5 (*fixed-argument strip checks*). *After relabeling so that $Q \leq M$, the Main Theorem holds for every fixed-argument strip $3 \leq Q \leq 12$.*

Proof. On $0 \leq u \leq D - 2Q - 6$, Lemma 4.4 applies. Lemma 3.6 places every gap cell in $u^2 \leq B_Q(D)$. Hence each certified pair satisfies

$$c_{j-2} u^{j-2} + c_j u^j = u^{j-2} (c_{j-2} + c_j u^2) \geq u^{j-2} (c_{j-2} + c_j B_Q(D)) \geq 0,$$

and the unpaired terms are nonnegative. Thus $S_Q \geq 0$, and positivity of the oriented denominator gives $T_s \geq 0$ throughout this strip.

For the top edge $v < 2Q + 6$, the inequality defining the flow region holds once

$$(D - 2Q - 5)^2 \geq (2Q + 1)(2D - 2Q + 1).$$

For $Q = 3, \dots, 12$, the corresponding threshold values are

$$31, 38, 45, 52, 59, 66, 73, 80, 86, 93.$$

Thus the largest onset is 93; all earlier edge cells are covered by Proposition 2.2. Corollary 3.4 applies off the gap. Applying the **fixed-argument-strips** verification for $Q = 3, \dots, 12$ proves the theorem. ■

Corollary 4.6 (*finite-family coverage*). *The Main Theorem holds whenever $\min(M, Q) \leq 12$, whenever $u \leq 14$, and for every cell with $D \leq 14826$. The first gap cell left after the fixed-offset and fixed-argument families is*

$$(D, Q, u) = (14827, 13, 15)$$

after relabeling so that $Q \leq M$.

Proof. The assertion for $\min(M, Q) \leq 12$ is Proposition 4.3 together with Theorem 4.5. The assertion for $u \leq 14$ is Proposition 4.1, Theorem 4.2, and Proposition 4.3.

It remains only to identify the first cell outside both finite families. For each D , heredity of the gap means that only the first parity-compatible $u \geq 15$ need be tested. At fixed (D, u) , the largest parity-compatible $x = D - 2Q$ with $Q \geq 13$ and

$$x^2 < (D + 1)^2 - (u + 1)^2$$

is the only candidate needed for the remaining strict bulk-failure inequality: if it fails, every smaller admissible x fails as well. Exact integer enumeration gives no candidate through $D = 14826$ and gives $(14827, 13, 15)$ at the next degree. The **fixed-argument-strips** generator records and repeats this census; the generic census rule is specified in Section 7. ■

5. Gap cells with even Q

Theorem 5.1 (*gap cells with even Q*). *After relabeling so that $Q \leq M$, every gap cell with Q even and $u \geq 15$ has $T_s \geq 0$.*

Proof. For each top-half index j , define

$$d_j := (D + 2)p_{j+1} - xp_j \quad (\text{deviation from the central rank-one relation}),$$

$$\Lambda := (D + 2)^2 - x^2 = 4(Q + 1)(M + 1) > 0, \quad \hat{u}_j := 2(j + 1) - D.$$

By Lemma 3.5, every positive parity-compatible offset between the center and the target remains in the gap; at the even- D center the same inequality below follows directly from $1 < (u + 1)^2 < U_-$. Thus, for each index j used,

$$(2j - D)^2 < (2j - D + 1)^2 < (D + 1)^2 - x^2.$$

We call the region $(2j - D)^2 < (D + 1)^2 - x^2$ the *oscillatory zone*. The display shows that all pointwise estimates for this zone used below hold along the entire induction path. We use five steps.

1. (*Turán-step identity*)

$$T_j = \frac{\widehat{u}_j \operatorname{Tur}(j)}{j+2} - \frac{p_{j+1}d_j}{(j+2)(D+1-j)}. \quad (5.1)$$

Since $\widehat{u}_j = 2(j+1) - D > 0$ at every top-half index used, $T_j \geq 0$ follows from $\operatorname{Tur}(j) \geq 0$ together with $p_{j+1}d_j \leq 0$.

2. (*positivity of the Turán determinant*)

$$(D+1-j) \operatorname{Tur}(j) = (D+1-j)p_j^2 - x p_j p_{j+1} + (j+1)p_{j+1}^2$$

is a quadratic form in (p_j, p_{j+1}) with discriminant $x^2 - [(D+2)^2 - (2j-D)^2]$ (using $4(j+1)(D+1-j) = (D+2)^2 - (2j-D)^2$), which is < 0 on the whole oscillatory zone $(2j-D)^2 < (D+1)^2 - x^2$. Therefore $\operatorname{Tur}(j) > 0$ there unless $(p_j, p_{j+1}) = (0, 0)$, which is impossible by Lemma 2.1. The leading coefficient $D+1-j$ is positive because $j \leq D$.

3. (*(p, d) system*) The first-order recurrence for (p_j, d_j) is

$$p_{j+1} = \frac{x p_j + d_j}{D+2}, \quad d_{j+1} = \frac{D-j}{(j+2)(D+2)} (x d_j - \Lambda p_j).$$

4. (*comparison map*) $\zeta_j := -d_j/p_j$ obeys

$$\zeta_{j+1} = \kappa_j \frac{\Lambda + x \zeta_j}{x - \zeta_j}, \quad \kappa_j = \frac{D-j}{j+2} \ (\leq 1 \text{ above the center});$$

the map is increasing in ζ on $[0, x)$ and in κ , and the $\kappa = 1$ map is exactly the tangent addition formula:

$$F(\sqrt{\Lambda} \tan a) = \sqrt{\Lambda} \tan(a + \theta^*), \quad \tan \theta^* = \frac{\sqrt{\Lambda}}{x}.$$

Reflection supplies the initial values (Q even here, $p_{D-j} = p_j$). For D even, $d_c = 0$ at $c = D/2$ (the rank-one relation of Theorem 3.2), so $\zeta_c = 0$; here $p_c \neq 0$, since otherwise the rank-one relation would contradict Lemma 2.1. For D odd, $p_{c+1} = p_c \neq 0$ at $c = (D-1)/2$ gives, in one exact step,

$$\zeta_{c+1} = \kappa_c (D+2-x) = \kappa_c \sqrt{\Lambda} \tan(\theta^*/2)$$

where $\tan(\theta^*/2) = \sqrt{\Lambda}/(D+2+x)$. Comparison with the majorant from the tangent addition formula then gives

$$\zeta_{c+\ell} \leq \sqrt{\Lambda} \tan((\ell - \delta/2)\theta^*), \quad \delta = D \bmod 2,$$

as long as the argument stays below $\pi/2$. More precisely, the path is $j = c, c+1, \dots, s$ when D is even and $j = c+1, c+2, \dots, s$ when D is odd. On this path the induction invariant is

$$p_j \neq 0, \quad 0 \leq \zeta_j \leq \sqrt{\Lambda} \tan((j - D/2)\theta^*) < x.$$

The displayed anchors establish its base case. If it holds at j , then $x - \zeta_j > 0$, so the (p, d) system makes p_{j+1} nonzero with the same sign as p_j ; monotonicity of the comparison map gives the next upper bound, while its positive numerator gives $\zeta_{j+1} \geq 0$.

5. (*angular bound*) On the gap, $(u+1)^2 \sin^2 \theta^* < 7/3$ because

$$\begin{aligned} (u+1)^2 \sin^2 \theta^* &\leq \frac{7}{6} \frac{D+3}{2Q+3} \frac{4(Q+1)(M+1)}{(D+2)^2} \\ &\leq \frac{7}{3} \frac{(M+1)(D+3)}{(D+2)^2} < \frac{7}{3}. \end{aligned}$$

The first line uses Lemma 3.6; the second uses $(Q+1)/(2Q+3) \leq \frac{1}{2}$, and the strict last inequality follows from $(D+2)^2 - (M+1)(D+3) = Q(D+3) + 1 > 0$. Since $x > 0$ in the gap, $\theta^* := \arctan(\sqrt{\Lambda}/x)$ lies in $(0, \pi/2)$. By Jordan's inequality ($\theta \leq \frac{\pi}{2} \sin \theta$ on $[0, \pi/2]$) and $u \geq 15$ (which gives $(u+2)/2 \leq \frac{9}{16}(u+1)$):

$$\frac{u+2}{2} \theta^* < \frac{9}{16} \frac{\pi}{2} \sqrt{\frac{7}{3}} < \frac{\pi}{2}.$$

Write $j = c + \ell$; then $2j - D = 2\ell - \delta$, and at the target $\ell - \delta/2 = u/2$. Therefore, at every step through $j = s$,

$$(\ell - \delta/2) \theta^* \leq \frac{u}{2} \theta^* < \frac{\pi}{2} - \theta^*.$$

Since $x = \sqrt{\Lambda} \cot \theta^* = \sqrt{\Lambda} \tan(\pi/2 - \theta^*)$, the comparison bound is strictly below x . This closes the induction invariant at every index through the target; in particular, no denominator in the Riccati map vanishes or changes sign. Consequently p retains its sign and $p_{j+1}d_j \leq 0$ from the first top-half cell through $j = s$. Steps (1) and (2) now give $T_s \geq 0$. $\blacksquare$

6. Gap cells with odd Q and proof assembly

6.1. Gap cells with odd Q

For each top-half index j with $p_j \neq 0$, define the scaled ratio

$$y_j := \frac{(D+2)p_{j+1}}{p_j}.$$

At a gap cell s with $p_s \neq 0$, the quadratic-form identity (3.1) gives

$$G(y_s)p_s^2 = (s+2)(D+1-s)(D+2)^2T_s,$$

where

$$G(y) = u(s+2)y^2 - x(u+1)(D+2)y + (u+2)(D+1-s)(D+2)^2. \quad (6.1)$$

Since $u \geq 1$, this is an upward parabola, and the gap inequality (3.7) gives two real roots $y_-(u) < y_+(u)$. We call the open interval between them the *gap window*. Thus $T_s \geq 0$ if and only if y_s avoids this interval.

For odd Q , the argument keeps the scaled ratio above the upper endpoint of the gap window. The next lemma gives the endpoint formula, obtained by the quadratic formula with $s = (D + r)/2$ in (6.1), and the comparison used to propagate this bound under the recurrence. The projective formulation in the proof of Theorem 6.2 also includes zero coefficients.

Lemma 6.1 (*odd-transition certificate*). *Let $D \geq 14827$, $Q \geq 13$, $u \geq 1$, $x = D - 2Q$, and suppose*

$$\frac{16(D+3)}{17} \leq x \leq D - 26, \quad 24u^2 \leq D + 3.$$

For each offset r for which the following square root is real, put

$$\begin{aligned} g_r &= \sqrt{x^2(r+1)^2 - r(r+2)(E - (r+1)^2)}, \\ y_+(r) &= \frac{(D+2)(x(r+1) + g_r)}{r(D+r+4)}, \quad \kappa(r) = \frac{D-r}{D+r+4}, \\ \Phi_r(y) &= (1 + \kappa(r))x - \kappa(r) \frac{(D+2)^2}{y}. \end{aligned} \tag{6.2}$$

If the windows at u and $u+2$ are both real, then

$$y \geq y_+(u) \implies \Phi_u(y) \geq y_+(u+2), \quad \Phi_u(\infty) = (1 + \kappa(u))x \geq y_+(u+2).$$

In addition, when D is odd,

$$\frac{(D+2)(2x+D+1)}{D+3} \geq y_+(1).$$

Proof. This is the exact polynomial certificate used in the induction for odd Q . Its symbolic obligations are as follows. Put

$$\begin{aligned} W_1 &= x(u+1) + g_u, \quad K = \frac{x((u+1)D + u^2 + 9u + 12)}{D + u + 4}, \\ B_0 &= u(D - u), \quad C_0 = (u+2)(D + u + 6), \\ \text{disc}_2 &= x^2(u+3)^2 - (u+2)(u+4)(E - (u+3)^2), \\ X_{\min 2} &= \frac{(u+2)(u+4)(E - (u+3)^2)}{(u+3)^2}, \quad X_{\max} = (D - 26)^2. \end{aligned}$$

Writing $g_{u+2} = \sqrt{\text{disc}_2}$, direct substitution gives

$$\begin{aligned} &(\Phi_u(y_+(u)) - y_+(u+2)) \frac{W_1 u(D + u + 4)(u+2)(D + u + 6)}{D + 2} \\ &= u(D + u + 4)(KW_1 - B_0 C_0 - g_{u+2} W_1). \end{aligned} \tag{6.3}$$

Every cleared factor is positive. Moreover $D > u$, $y_+(u) > 0$, and

$$\Phi'_u(y) = \kappa(u) \frac{(D+2)^2}{y^2} > 0,$$

so it is enough to certify this boundary value.

After reducing g_u^2 to its defining polynomial, clear the denominator of K and define P_1, R_1 uniquely by

$$(D+u+4)^2 [(KW_1 - B_0C_0)^2 - \text{disc}_2 W_1^2] = 4(u+2)(P_1 + xR_1g_u);$$

the left-hand side is a polynomial because $(D+u+4)K$ is, and the clearing factor is a positive square, so no sign conclusion below is affected. As polynomials in $X = x^2$, P_1 is concave, with leading coefficient $2(u+1)^2(u-D)(D+u+6) < 0$, and R_1 is decreasing, with coefficient $2(u+1)(u-D)(D+u+6) < 0$. The replay certifies

$$\begin{aligned} O_1 &= R_1(X_{\max}) \geq 0, \\ O_2 &= P_1(X_{\max}) \geq 0, \\ O_3 &= (u+3)^4 P_1(X_{\min 2}) \geq 0, \\ O_4 &= \frac{(D+u+4)^2}{u+2} (K^2 - \text{disc}_2) \Big|_{X=X_{\max}} \geq 0. \end{aligned}$$

The reality of the successor window and the bound on x place $X = x^2$ in $[X_{\min 2}, X_{\max}]$. For $u \geq 25$, the four obligations follow by coefficient inspection after $D = 24u^2 - 3 + \rho$, $u = z + 25$; for $1 \leq u \leq 24$, they follow after $D = 14827 + \rho$ (here $\rho, z \geq 0$ are slack variables). Thus concavity gives $P_1 \geq 0$ at both ends of the allowed X -interval and hence throughout it, while monotonicity gives $R_1 \geq 0$. Next, $Kx(u+1)$ is increasing in X , because its numerator $X(u+1)((u+1)D + u^2 + 9u + 12)$ has positive coefficients; the same numerator shows $K > 0$. At the gap boundary $X = u(u+2)(E - (u+1)^2)/(u+1)^2$ the value of $Kx(u+1) - B_0C_0$ is $4u(D+3)(u+2)^2/(u+1)$; equivalently, after multiplication by the positive factor $(u+1)(D+u+4)/(u(u+2))$, it is exactly $4(D+3)(u+2)(D+u+4)$. The strict gap inequality therefore gives $Kx(u+1) - B_0C_0 > 0$, so the squared inequality has the correct unsquared sign and proves the implication for finite y . Finally, after multiplication by the positive factor in O_4 , $K^2 - \text{disc}_2$ decreases linearly with X , with coefficient $4(u-D)(D+u+6) < 0$. Thus O_4 and $K > 0$ prove the state at infinity.

For the odd- D base, clearing the radical makes the claimed inequality a concave quadratic in x . Its values at $x = 16(D+3)/17$ and $x = D - 26$ become coefficient-positive polynomials after $D = 14827 + \rho$. All reductions are identities in $\mathbb{Z}[D, u, X]$ and are replayed as described in Section 7.2. ■

Theorem 6.2 (*gap cells with odd Q*). *After relabeling so that $Q \leq M$, every gap cell with Q odd has, for either parity of D , $T_s \geq 0$.*

Proof. By Theorem 4.2 and Corollary 4.6 we may assume $Q \geq 13$, $u \geq 15$, $D \geq 14827$. First record the bounds needed by the transition certificate. Lemma 3.6 gives

$$(u+1)^2 \leq \frac{7}{6} \frac{D+3}{29}, \quad 24u^2 \leq \frac{168}{174} (D+3) < D+3.$$

Also $x = D - 2Q \leq D - 26$. Since $U = (u + 1)^2 \geq 256$ and $U \leq (D + 3)/24$, the gap inequality gives

$$\begin{aligned} x^2 &> \left(1 - \frac{1}{U}\right)(E - U) > \frac{8}{9} \left(E - \frac{D + 3}{24}\right) \\ &= E \frac{24(D + 3) - 1}{27(D + 3)} \geq E \frac{256}{289}, \end{aligned}$$

where the last inequality is equivalent to $24(D + 3) \geq 289$. Hence

$$17x > 16(D + 3).$$

For a top-half index j , write $r = 2j - D$. By Lemma 3.5, every positive same-parity offset below the target has a real gap window, so g_r is real on the induction path. To include zeros without division, define the projective state

$$\mathcal{C}_r := \left\{ (a, b) : a \neq 0, \frac{(D + 2)b}{a} \geq y_+(r) \right\} \cup \{(0, b) : b \neq 0\}.$$

The second set is the single state $y = \infty$, and simultaneous zeros are excluded by Lemma 2.1. On finite states the recurrence induces the Möbius map

$$y_{j+1} = (1 + \kappa_j)x - \kappa_j \frac{(D + 2)^2}{y_j}, \quad \kappa_j = \frac{D - j}{j + 2},$$

increasing in y on $y > 0$.

By the window interpretation in (6.1), it is enough to prove $(p_j, p_{j+1}) \in \mathcal{C}_{2j-D}$ from the first noncentral cell to s .

Reflection for odd Q is $p_{D-j} = -p_j$. For D even: $p_c = 0$ at $c = D/2$ and $p_{c+1} \neq 0$ by Lemma 2.1, and $y_{c+1} = 2(D + 2)x/(D + 4)$ exactly. For D odd: $p_c \neq 0$ because $p_{c+1} = -p_c$, and hence $y_c = -(D + 2)$ at $c = (D - 1)/2$, so $y_{c+1} = (D + 2)(2x + D + 1)/(D + 3)$ exactly. For D even, the base inequality is $y_{c+1} \geq y_+(2)$; eliminating the square root reduces it to $x^2 \leq (D + 4)^2$, which always holds. For D odd, the base inequality is $y_{c+1} \geq y_+(1)$, the final assertion of Lemma 6.1.

At each subsequent offset, Lemma 6.1 maps both the finite and infinite parts of $\mathcal{C}_r$ into $\mathcal{C}_{r+2}$. Induction therefore gives $(p_j, p_{j+1}) \in \mathcal{C}_{2j-D}$ along the induction path. At the target, either $p_s = 0$ and (3.1) gives $T_s = up_{s+1}^2/(D + 1 - s) \geq 0$, or $y_s \geq y_+(u)$ and hence $G(y_s) \geq 0$. Thus $T_s \geq 0$. Cells with $D \leq 14826$, or $Q \leq 12$, or $u \leq 14$ were already covered by Corollary 4.6. $\blacksquare$

6.2. Assembly: proof of the Main Theorem

Proof of Theorem 1.1 and Corollary 1.2. Fix $M, Q \geq 0$, put $D = M + Q$, and choose $D/2 \leq s \leq D$. By swapping the exponents if necessary, assume $x = M - Q \geq 0$ (Section 1.5). Write $u = 2s - D$ and $U = (u + 1)^2$.

1. If $s = D$: $T_D = p_D^2 = 1 > 0$.

2. If $u = 0$: Theorem 3.2.
3. If $x^2U \leq (U - 1)(E - U)$: Theorem 3.1 (bulk).
4. If $U \geq (D + 1)^2 - x^2$: Theorem 3.3 (flow).
5. Otherwise the cell is a gap cell (Corollary 3.4). After relabeling, $Q \leq M$:
 - $u \leq 3$: Proposition 4.1;
 - $Q \leq 2$: Proposition 4.3;
 - $4 \leq u \leq 14$: Theorem 4.2;
 - $3 \leq Q \leq 12$: Theorem 4.5;

If none of these cases applies, then $u \geq 15$ and $Q \geq 13$. Corollary 4.6 also gives $D > 14826$. Theorem 5.1 applies when Q is even, and Theorem 6.2 applies when Q is odd.

Every branch ends in $T_s \geq 0$. The corollaries follow by telescoping from $\text{Tur}(D) = 1$ and by the reflection symmetry of Tur (Section 1.1). ■

7. Computer-assisted verification

7.1. Methodology

The proof is computer-assisted in the following precise sense. The symbolic identities and finite positivity claims used in Sections 2–6 have exact-arithmetic replays: rational arithmetic (SymPy [MSP⁺17]) for symbolic identities, integer Sturm chains for univariate positivity claims, and GMP integers for the exhaustive scan. No floating-point comparison or unquantified asymptotic estimate is used to prove the Main Theorem. In particular, the orbit tangent-bound comparison in the verifier for even Q is exact: every orbit angle is a nonnegative multiple of θ^* (plus $\theta^*/2$ for odd D), where the tangent is a rational multiple of $\sqrt{\Lambda}$, and the tangent addition formula preserves that form. The verification patterns are:

- *coefficient-positive polynomials*: after explicit integer substitutions (e.g. $D = 24u^2 - 3 + \rho$, $u = z + 25$ in Lemma 6.1) the certified polynomials have all coefficients non-negative, so nonnegativity follows by inspecting the coefficients;
- *Sturm counts* on explicitly given intervals for the univariate cases (fixed-argument pairing thresholds, interval base case, finite strips);
- *threshold censuses*: because the gap is $U < U_-$, only the first parity-compatible offset above a declared cutoff need be tested. At fixed (D, u) the admissible values $x = D - 2Q$ have the parity of D , and the two gap inequalities are exactly

$$\frac{(U - 1)(E - U)}{U} < x^2 < (D + 1)^2 - U.$$

Thus the largest parity-compatible x below the upper bound and the prescribed Q cutoff decides existence exactly;

- *exhaustive exact scans* for the remaining finite regions ($M, Q \leq 90$; $D \leq 1200$, 289,261,901 cells), through the certified recurrence, with divisibility checked before every GMP exact division. The resulting values are checked against the defining binomial convolution for every coefficient with $D \leq 12$ and at deterministic indices in six larger rows up to $D = 1200$. The cell count is asserted in-program against

$$\sum_{D=0}^{1200} (D+1)(\lfloor D/2 \rfloor + 1) = 289,261,901;$$

- *independent exact replays*: a standalone C++ checker implements rational and sparse-polynomial arithmetic, Euclidean division, Sturm chains, modular no-root tests, and Sylvester determinants over Boost integers. It checks the exported fixed-argument fractions on degree-complete grids, reconstructs the finite-offset quadratic forms from the fraction-free recurrence, multiplies their factorizations, repeats the sector bounds, enforces each resultant's degree bound, and verifies it at one more integer point than that bound. A second finite scanner constructs each row from both Pascal parents $(1+w)p(M-1, Q)$ and $(1-w)p(M, Q-1)$ rather than from the three-term recurrence, and repeats all 289,261,901 cells;
- *end-to-end consistency checks* confirming that the declared regions cover all cells with $D \leq 90$ and that each argument's output matches direct evaluation at explicit cells.

The source programs specify the proof obligations and their exact checks; the committed text files record deterministic replay results, with elapsed times excluded. The manifest identifies the corresponding source and log files. Hashes check snapshot integrity; assessing a computer-assisted step also entails inspecting or replaying the program that produces its **PASS** line. The finite-offset log records, for every offset and parity, the factor-degree pattern, domination threshold, resultant degree and integer-candidate count, and the number of exact spot cells tested in each reflection parity. The fixed-argument log records every paired coefficient index and its certified onset. The verifier for odd Q checks the fully cleared transition identity (6.3) before any endpoint positivity test.

The certificate-generating sources form part of the argument. Their software dependencies and the scope of the independent implementation are described in Section 7.3. The file **certificates/CERTIFICATE_MAP.md** maps every computer-assisted theorem to its generator, exact log, proof obligations, and replay command. The companion **referee checklist** gives a function-by-function inspection route: it identifies the constructor, sign test, nondegeneracy test, threshold census, and final acceptance predicate used by each verifier. This separates review of the mathematical implication from the diagnostic examples printed by the programs. The additional **certificates/INDEPENDENT_CHECKS.md** specifies the second implementation and its mutation tests, while **certificates/INDEPENDENT_REPLAY_SIGNOFF.md** provides an external replay form.

7.2. Certificate specifications

For clarity, we state here the elementary certificate rules used by the programs. They reduce the computer-assisted portions to finite lists of integer/rational arithmetic assertions.

Lemma 7.1 (*sign-certificate rules*). *The following are valid exact certificates.*

1. Let $f(M, Q)$ have total degree d , put $t = Q/M$, and write

$$M^{-d}f(M, Mt) = L(t) + \sum_{\nu \geq 1} M^{-\nu} L_\nu(t).$$

On $0 \leq t \leq t_0$, let m_L be the minimum of L over the points of a mesh of width t_0/N , and suppose $|L'| \leq B$. Then $L(t) \geq m_L - Bt_0/(2N) =: \ell$. If $|L_\nu(t)| \leq S_\nu$ and $\sum_{\nu \geq 1} S_\nu M^{-\nu} < \ell$, then $f(M, Q) > 0$.

2. If $F \in \mathbb{Z}[D]$ has positive leading coefficient, has no real root in $[D_0, \infty)$ by its exact Sturm chain, and $F(D_0) > 0$, then $F(D) > 0$ for all $D \geq D_0$.
3. After clearing nonzero rational content, if $f, g \in \mathbb{Z}[M, Q]$ have a common integer zero, then its M -coordinate is an integer root of $\text{Res}_Q(f, g)$. Factoring this univariate resultant over $\mathbb{Z}$, checking every linear-factor root, and then computing the specialized gcd in $\mathbb{Z}[Q]$ is therefore exhaustive.

Proof. Part (1) is the mean-value theorem on the nearest mesh point followed by the triangle inequality. Part (2) is the intermediate-value theorem together with the Sturm root count. Part (3) is the defining vanishing property of the resultant; every integer root of a univariate integer polynomial occurs as a rational linear factor. ■

Finite offsets. For each $u \in \{4, \dots, 14\}$ and $\varepsilon \in \{\pm 1\}$, the inputs are defined recursively by the scaled chain in Theorem 4.2, followed by

$$(\lambda, \mu) = \text{coeff}_{a,b}(P_u - \varepsilon c_u a), \quad G = \mathcal{A}\mu^2 - \mathcal{B}\lambda\mu + \mathcal{C}\lambda^2.$$

The program constructs these objects in $\mathbb{Q}[M, Q]$, verifies both identities in (4.1), removes $g = \text{gcd}(\lambda, \mu)$, and applies Lemma 7.1(1) with $N = 512$ and $t_0 = (u+1)^{-2}$. The core factor degrees and largest domination onsets are:

u	4	5	6	7	8	9	10	11	12	13	14
$\text{deg}(+)$	4	4	6	6	8	8	10	10	12	12	14
$\text{deg}(-)$	2	4	4	6	6	8	8	10	10	12	12
M_0	4	9	13	25	33	50	63	85	103	130	153

Here $\text{deg}(\pm)$ refers to $\varepsilon = \pm 1$; the alternate pinning has the same core degree. The committed log gives every factor multiplicity, resultant degree, and integer-candidate count, while the source enumerates and checks the complete candidate set and specialized gcds prescribed by Lemma 7.1(3). The verifier also repeats, as an exact heredity-reduced integer scan, the census fact that no gap cell with $u \geq 4$ and $Q \geq 3$ has $D < 442$ (the input to the sector bound $M \geq 425$), and evaluates each constrained identity at the first gap cells of both reflection parities, reporting the tested cell counts.

Fixed arguments. The finite binomial-ratio input and denominator orientation are stated in Lemma 4.4. After expansion, for even Q the partner indices $j - 2$ are $4r, 4r + 1$ for $0 \leq r < Q/2$; for odd Q , they are $4r + 2, 4r + 3$ for $0 \leq r < (Q - 1)/2$. Lemma 7.1(2) certifies all coefficient signs and $c_{j-2} + c_j B_Q(D) > 0$ from the common onset $D_0(Q) = 2Q + 6$; every unpaired coefficient is certified positive on the same ray. This is the complete pairing list for $3 \leq Q \leq 12$.

Endpoints for odd Q . The four polynomial obligations in Lemma 6.1 are, in the notation there,

$$\begin{aligned} O_1 &= R_1(X_{\max}), & O_2 &= P_1(X_{\max}), \\ O_3 &= (u + 3)^4 P_1(X_{\min 2}), & O_4 &= (D + u + 4)^2 \frac{K^2 - \text{disc}_2}{u + 2} \Big|_{X=X_{\max}}. \end{aligned}$$

After $D = 24u^2 - 3 + \rho$, $u = z + 25$, their respective term counts and least coefficients are

	O_1	O_2	O_3	O_4
# terms	24	48	77	30
min coefficient	2	58	1	1

and hence all four are nonnegative for $u \geq 25$. For $1 \leq u \leq 24$, substituting $D = 14827 + \rho$ gives nonnegative coefficients directly; no Sturm fallback is used. These counts and minima are recomputed, not hard-coded, by the verifier for odd Q .

7.3. Code availability and software dependencies

This manuscript is Version 1.0.9. Its source and PDF are maintained in the [project repository](#), where `certificates/MANIFEST.sha256` identifies the manuscript, verification programs, and deterministic outputs byte-for-byte. The verification programs and logs are unchanged from the [version 1.0.6 release](#) [Pol26], which includes the source archive, hash records, and replay profiles. GitHub is the distribution site. No external human replay attestation is claimed for this version.

To avoid a circular identifier (the full manifest includes the manuscript), `certificates/PAYLOAD.sha256` separately hashes only the proof generators, exact logs, dependency locks, license, certificate map, and referee checklist. Its own SHA-256 digest is

0856e33af616ca7883835a449e2e1ee2b9a28c91bdc946ed5a2e752b47e02681

This printed digest identifies the accompanying verification files; the full manifest additionally binds the article and the PDF built in the reference toolchain.

The verification code is available under the BSD 3-Clause License. The manuscript and explanatory documentation are available under the Creative Commons Attribution 4.0 International License; the repository's `LICENSE` file states the component boundaries.

The primary implementation uses Python 3, SymPy 1.12, a C++17 compiler, and GMP; its symbolic operations include factorization, gcd, resultants, and Sturm root counting. The independent implementation uses the same compiler and checks the decisive outputs with repository-local polynomial algorithms and Boost multiprecision integers. Both implementations depend on their source code, compiler or runtime, and integer primitives. Independence here refers to the implementation, not to replication by another researcher.

Reference software versions and individual commands are recorded in [paper/README.md](#). The targets `payload-check` and `verify` check stored hashes; `replay-all` executes the complete set of proof programs and independent checks. The README also describes the shorter audit targets, the toolchain report, and resource profiling. Resource values are machine-dependent and are therefore excluded from the deterministic proof logs.

7.4. Verification inventory

The first column gives a verification task or log stem; each link opens the corresponding committed log in `certificates/`. The exact program paths, outputs, and replay commands are listed in [CERTIFICATE_MAP.md](#). In particular, the two scan logs come from [cpp/exhaustive-turan-scan.cpp](#) and [cpp/independent-pascal-scan.cpp](#).

Verification task / log	Checks
recurrence-and-small-scan	Recurrence identity ($M, Q \leq 30$) and $T_s \geq 0$ exhaustively for $M, Q \leq 90$.
exhaustive-scan-D1200	Exact scan of all 289,261,901 cells with $D \leq 1200$; divisibility tests and convolution audits (Proposition 2.2).
independent-scan-D1200	The same finite set by Boost/Pascal construction; agreement of both parents, reflection, and endpoint checks.
regime-decomposition	Theorems 3.1–3.3, Corollary 3.4, Proposition 4.1: identities, grids, coverage for $D \leq 90$, and gap census.
finite-gap-offsets	Theorem 4.2: $4 \leq u \leq 14$, both parities; sector estimate and threshold $D = 4586$.
small-argument-cases	Proposition 4.3: $Q \leq 2$ for all D .
fixed-argument-strips	Theorem 4.5 and Corollary 4.6: $3 \leq Q \leq 12$; threshold census $D = 14826$.
even-minimum-gap	Theorem 5.1: identities, angular bound, and end-to-end checks for even Q and both parities of D .
odd-minimum-gap	Theorem 6.2: interval algebra, coefficient signs, and recurrence checks near $D = 14828$.
independent-certificate-check	Separate replay of ratios, Sturm chains, factors, sector bounds, resultants, and endpoint signs. The accompanying mutation-tests log records rejection of all twelve mutations.

Table 2. Verification inventory.

8. Related work

The classical Turán inequality for Legendre polynomials, discussed by Szegő [Sze48], is $P_n(x)^2 - P_{n-1}(x)P_{n+1}(x) \geq 0$ on the orthogonality interval; its descendants form a large literature on the *positivity* of Turán determinants as functions of the argument x , with the

degree index n fixed, for broad classes of orthogonal polynomials (for general background see [Sze75]). The following directions are closest to the present theorem.

Recurrence-based positivity criteria. Szwarc [Szw98] gave general criteria (monotonicity conditions on the three-term recurrence coefficients) under which orthogonal polynomials satisfy the Turán inequality. Bustoz and Ismail [BI97, Section 2, Theorem 1, (2.3), p. 2] gave weighted sum-of-squares representations for Turán determinants arising from symmetric three-term recurrences. Berg and Szwarc [BS09] proved two-sided bounds on normalized Turán determinants, and Szwarc [Szw21] extended the positivity criteria further. Krasikov [Kra11] developed Turán inequalities adapted to three-term recurrences with monotone coefficients. Recent general criteria and nonnegative representations are developed in [Kah26]; that work also propagates positivity between determinant indices under recurrence hypotheses. These results concern the *sign* or propagated sign of Tur under prescribed recurrence and normalization hypotheses. Here the comparison is $\text{Tur}(s) \geq \text{Tur}(s+1)$ in the ordinary coefficient normalization. The integer arguments considered here belong to the Krawtchouk orthogonality support. Section 1.4 explains the effect of degree-dependent rescaling on this comparison.

Krawtchouk bounds. Krasikov (partly with Litsyn) developed Turán-type and Laguerre-type inequalities for discrete orthogonal polynomials and used them for sharp, parameter-uniform bounds on binary Krawtchouk polynomials, their extreme zeros, and Christoffel–Darboux kernels [KL96, Kra01, Kra03, Kra05]. The inequality recorded in [Kra05, Theorem 5, (18), p. 79] gives a fourth-degree refinement coupling two adjacent Turán determinants. Krasikov credits this refinement to Mařík and its extension to the full Laguerre–Pólya class to Dimitrov. The generating-function framework includes binary Krawtchouk polynomials and is written for exponential coefficients $\sum_k u_k z^k / k!$; converting it to the ordinary generating coefficients in (1.1) introduces a degree-dependent factorial normalization. Its conclusion concerns positivity and a product of adjacent determinants; Theorem 1.1 concerns the ordered difference $\text{Tur}(s) - \text{Tur}(s+1)$ in the ordinary coefficient normalization.

Computer algebra proofs. Gerhold and Kauers [GK06, Section 2] describe a computer algebra proof of a Turán inequality using induction and cylindrical algebraic decomposition. The present argument combines recurrence comparisons with exact polynomial certificates and a finite integer-arithmetic scan.

Specialized Schur functions and character values. Proposition 1.4 identifies $\text{Tur}(s)$ with the rectangular character value at the diagonal matrix $\text{diag}(I_M, -I_Q)$. The factorization theorem of Littlewood and Prasad, presented in [AK22, arXiv version 3, Theorem 2.5], treats general linear characters under balanced root-of-unity specializations. Karmakar [Kar25, arXiv version 1, Theorem 4.1] gives formulas at arbitrary order-two conjugacy classes; outside the factorized cases these formulas are alternating sums of products of dimensions. These works give formulas for individual character values. Here we study their positivity and ordering as the rectangle height varies. Reiner and Stanton [RS98] prove symmetry and unimodality for certain differences of *principally* specialized Schur functions $s_\lambda(1, q, \dots, q^n)$. Their specialization and varying shapes differ from the fixed $\{1, -1\}$ alphabet and rectangle-height sequence here. These works place the Main Theorem in the context of character specialization and Schur unimodality as well as classical Turán inequalities.

Coefficient and asymptotic inequalities. Turán inequalities for coefficient sequences and Jensen polynomials have a separate classical theory, for example Craven–Csordas [CC89]. Uniform asymptotics for Krawtchouk polynomials in the oscillatory regime are also available [LW00]. The cited results address coefficient inequalities and asymptotic expansions; the present theorem concerns degree-index monotonicity at a fixed integer Krawtchouk argument.

Coding-theoretic Krawtchouk analysis. Binary Krawtchouk polynomials at integer arguments are the spectral kernel of the Hamming scheme [Del73, Lev95, MS77]; positivity of Krawtchouk quadratic expressions at integer points is central to linear-programming bounds.

To our knowledge, Theorem 1.1 has not previously been stated for the coefficient normalization of $(1+w)^M(1-w)^Q$ at integer (M, Q) , equivalently for the ordered rectangular character values in (1.3).

9. Further questions

1. **Uniform- Q pairing structure.** The fixed-argument strip certificates (Theorem 4.5) exhibit a recurring pattern: after removing positive content, the negative coefficients of S_Q are paired with positive neighbors two degrees below. With M fixed, write $p_s^{(Q)} = [w^s](1+w)^M(1-w)^Q$. Can the contiguity recursion $p_s^{(Q+1)} = p_s^{(Q)} - p_{s-1}^{(Q)}$ be used to propagate the coefficient-pairing inequalities of Theorem 4.5, with their interval bounds, to all Q ? Such a propagation would treat the entire gap uniformly in Q and yield another proof of the Main Theorem.
2. **Beyond the symmetric weight.** For which $a, b > 0$, with $a \neq b$, and which coefficient normalization does an analogous Turán-step monotonicity statement hold for $p_s(a, b) = [w^s](1+aw)^M(1-bw)^Q$ for all integers $M, Q \geq 0$? More generally, which conditions on the locations and multiplicities of positive and negative real roots permit such a statement for other real-rooted products? The central reflection constraints used here depend on the symmetric two-root family; which parts of the argument extend beyond it?
3. **Combinatorial meaning of $\text{Tur}(s) \geq 1$.** Corollary 1.2 gives $\text{Tur}(s) \geq 1$ for every $0 \leq s \leq D$. Is there a lattice-path or coding-theoretic injection realizing $p_s^2 - p_{s-1}p_{s+1} - 1$ as a counting quantity at integer arguments?
4. **Equality away from the center.** Theorem 3.2 classifies central equality, while the proof of the Main Theorem does not track equality through every certificate branch. Are there any noncentral cells with $T_s = 0$?

AI-use disclosure

The proof architecture was developed with assistance from GPT-5.6 Sol. The system was used as an interactive research assistant to develop and stress-test the argument, audit citations and calculations, prepare exact verification code, and revise the manuscript. The author directed the research, selected and checked the final arguments, and takes responsibility for all mathematical claims, citations, computations, and wording.

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